

Project Coversheet

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Project Title (Example – Week1, Week2, Week3, Week 4)	Churn Prediction for StreamWorks Media

YOU CAN START YOUR PROJECT FROM HERE

Introduction

I have joined the data strategy team at StreamWorks Media who are a fast-growing UK-based video streaming service competing with global companies such as Netflix and Amazon Prime. With customer acquisition becoming more expensive and competition increasing I have been tasked by manager to analyse customer churn which are users who cancel their subscriptions. The goal is to understand churn drivers, pattern recognition and more. Analysis includes:

- Data Loading and Cleaning
- Statistical Analysis
- Visual and Correlation Analysis
- Data Engineering
- Churn modelling with logistic regression
- Business recommendation based on patterns and findings

Data Cleaning Summary

During this process many things were handled and performed to prepare the dataset for analysis and modelling, things including correlations, missing data handled and encoding categorical features.

- I converted `signup_date` and `last_active_date` into a datetime format using `pd.to_datetime()`
- Numerical fields like `age`, `monthly_fee`, `complanits_raised` and others were stored as numeric types instead of string types.
- Binary fields like `is_churned` were converted into encoded formats

Missing values were handles as fields like `is_churned` and `average_watch_hours` were removed. Numeric colums with missing values like `age` were filled using median. Categorical fields with missing values were filled with unknown values, after cleaning data no missing values was confirmed with the string `.isnull().sum()`.

For encoding categorical variables, several steps were applied optional commands such as `pd.get_dummies()` or `LabelEncoder` were used. I went with `pd.get_dummies()` and used it for values such as:

- Gender
- Country
- Subscription type
- Received promotions
- Referred to by friend
- Age group

These categorical values were converted into numeric form.

Feature Engineering Summary

During this stage several new features were added/created to support analysis and predictive modelling.

The first new feature created was `tenure_days` which showed the number of days between `signup_date` and `last_active_date`. This measured how long a user had been active for which can be useful for analysing customer loyalty.

The second new feature created was `is_loyal` which is a binary feature with numbers such as 1 showing users with more than 180 days of tenure and 0 showing users with less than 180 days of tenure. The purpose of this addition is to help identify long-term engagement users.

The third new feature was `watch_per_fee_ratio` which measures perceived value and how much content a user consumes relative to the prices they pay with relating fields from `average_watch_hours` and `monthly_fee`

The fourth new feature was `heavy_mobile_users` consisting of numeric values like 1 which showed `mobile_app_usage_pct > 70%` else 0. This indicates strong preference for mobile usage which may affect churns.

`Age_group` was created which grouped users into categories for example 25-34, this allows churn and engagement analysis by each demographic group.

One-Hot encoded variables were transformed into dummy variables using the command `pd.get_dummies()`:

- Gender
- Country
- Subscription_type
- Received_promotions
- Referred_by_friend
- Age_group

Key Findings

Chi-square tests were done to examine whether churn was associated with categorical factors.

- Gender vs Churn showed males, females churn at similar rates and was no statistically significant relationship
- Received promotions vs churn showed users who received promotions churn less than those who don't
- Referred by friend vs churn shows referral status does not appear to influence churn outcomes.

There is no impactful difference in watch time between churned and retained customers.

Correlations between numerical values and churn were all quite small including things like `tenure_days` and `complaints_raised`.

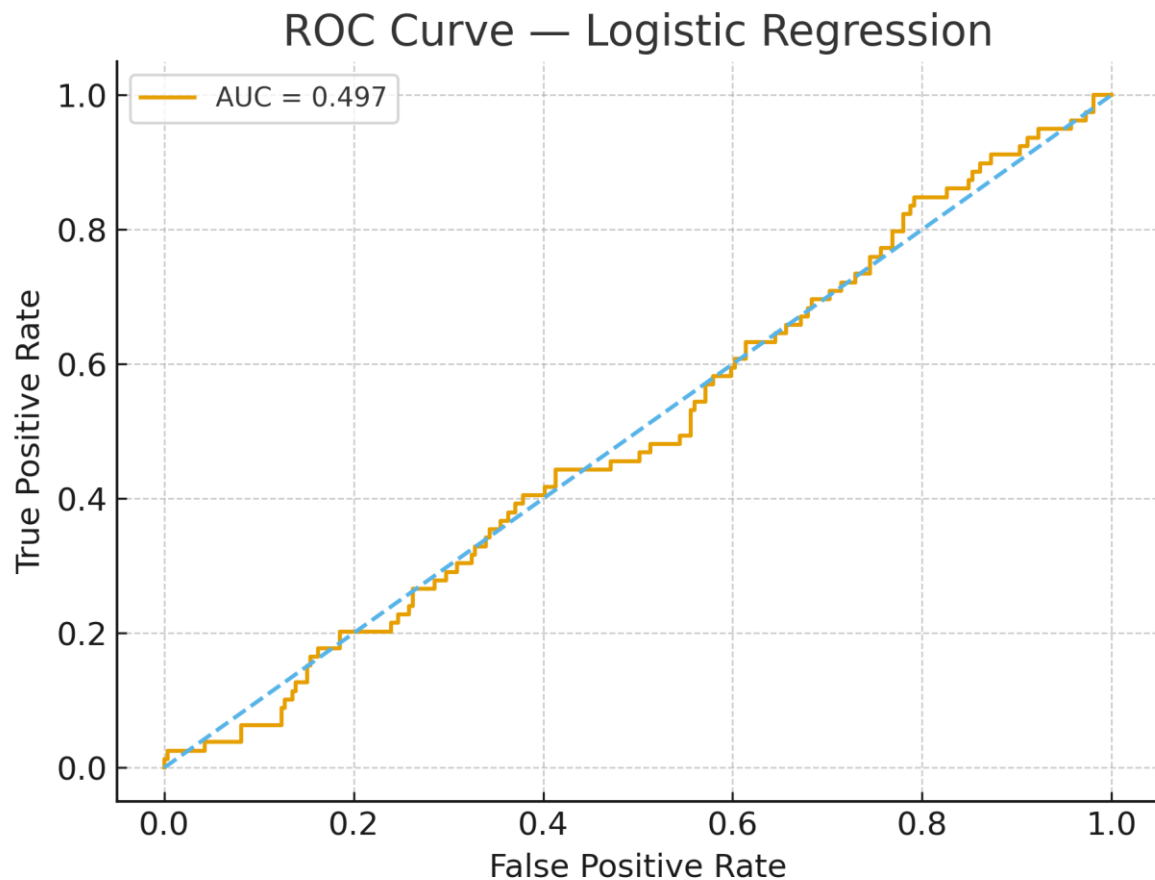
Loyalty correlates with engagement, users tenure over 180 days (`is_loyal = 1`) were engaged, less likely to churn

Promotions may help retention even though churn was lower among users who had promotions.

Model Results

Logistic regression model was trained to predict whether a user churned.

- Accuracy appears high because most users did not churn and the model predicted almost all users as not churn
- Precision = 1.00 meaning single user predicted as churn actually churned
- Recall = 0.013 meaning the model captured only 1.3% of churners
- AUC = 0.50 indicates the model has no power, performing no better than random



The diagonal line represents random guessing

The model does not meaningfully distinguish churners from non-churners

Confirming the models weakness is predictive power

Even though prediction is weak, logistic regression still gives directional insights

Higher age groups mean higher churn; older customers may find the interface hard or less appealing

Targeted ui/ux design principles may help

Watch_per_fee_ratio means lower churn because customers who get good or better value stay longer

Personalized content may increase perceived value

Received_promotions means lower churn as incentives help retain price sensitive users

Targeted programs could help retention

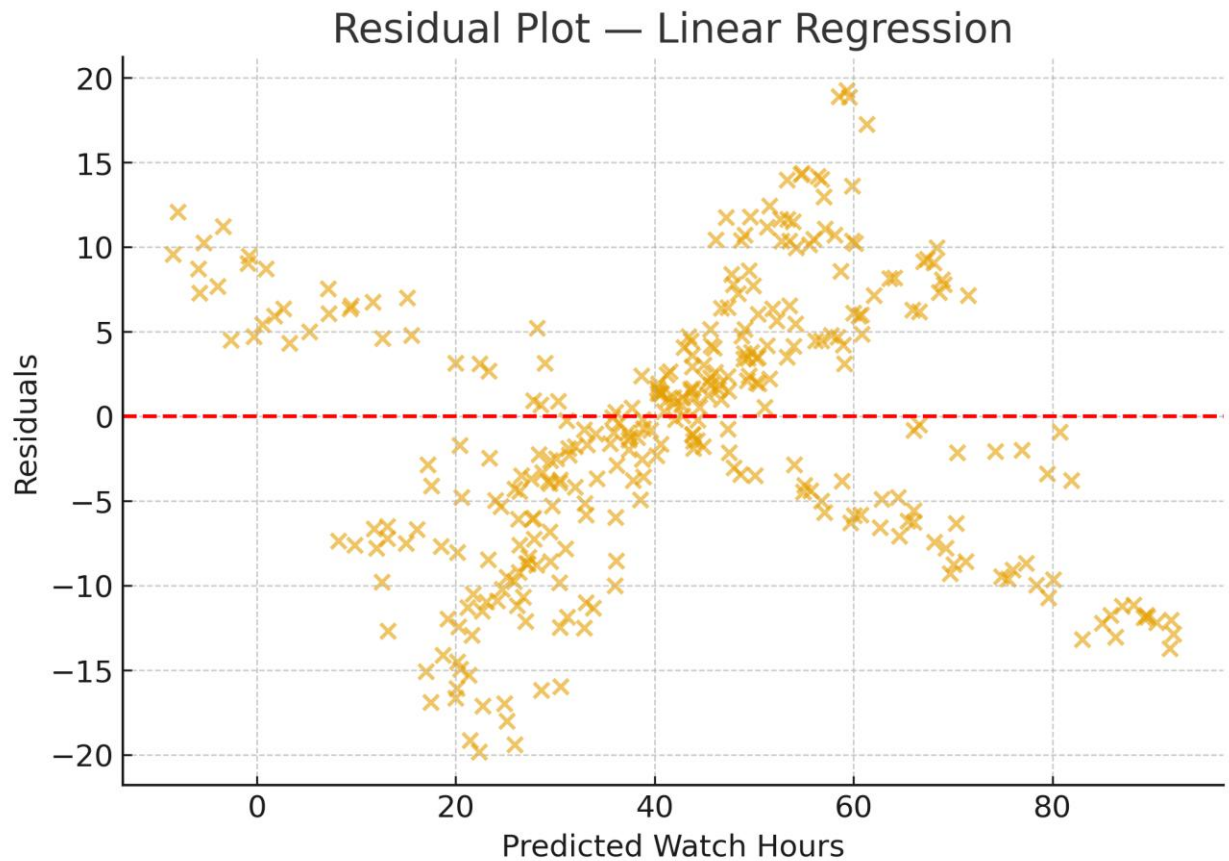
Predicting watch hours with linear regression

High RMSE and MAE show poor predictive accuracy

None of the available features effectively explain watch time

Performance is low as influenced by day-to-day usage patterns including

- Number of sessions
- Recency
- Content categories watched
- Time of day
- Recommendations



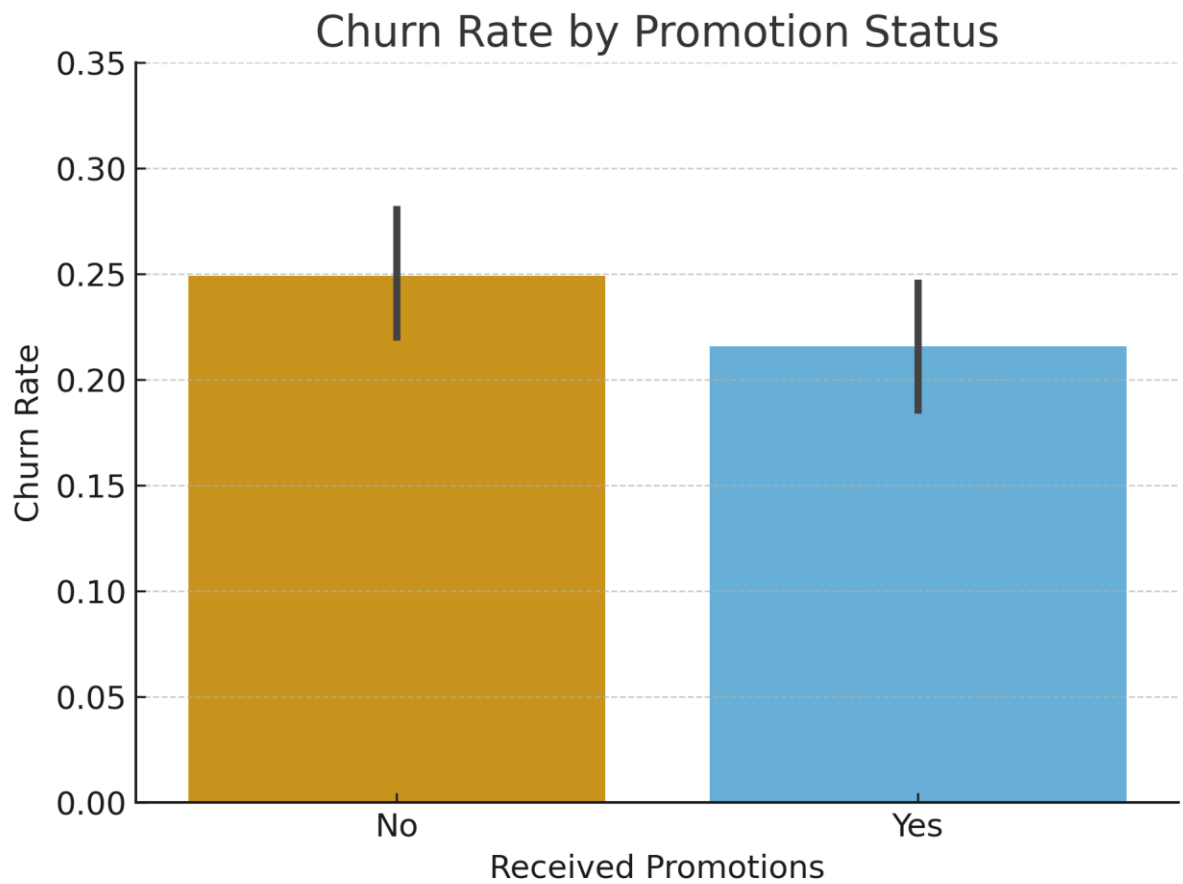
Residuals appear widely scattered around with no visible pattern

The linear regression models does not meaningfully predict watch time

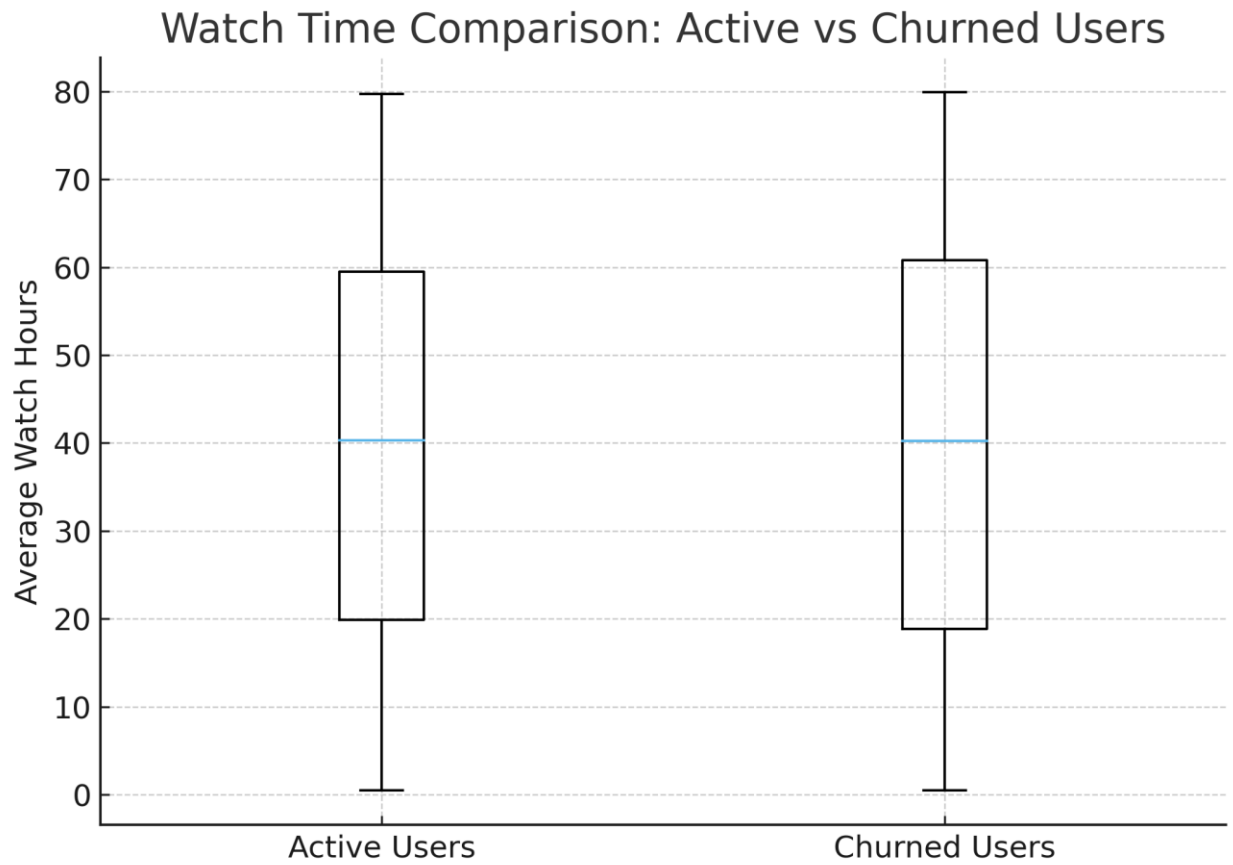
The top 3 predictors of watch time are

- Monthly fee which monitors value perception for premium tiers
- Country meaning local content strategies or regional preferences drive watch time
- Heavy_mobile_users meaning mobile-first experience is important for optimizing recommendations and playback Ux.

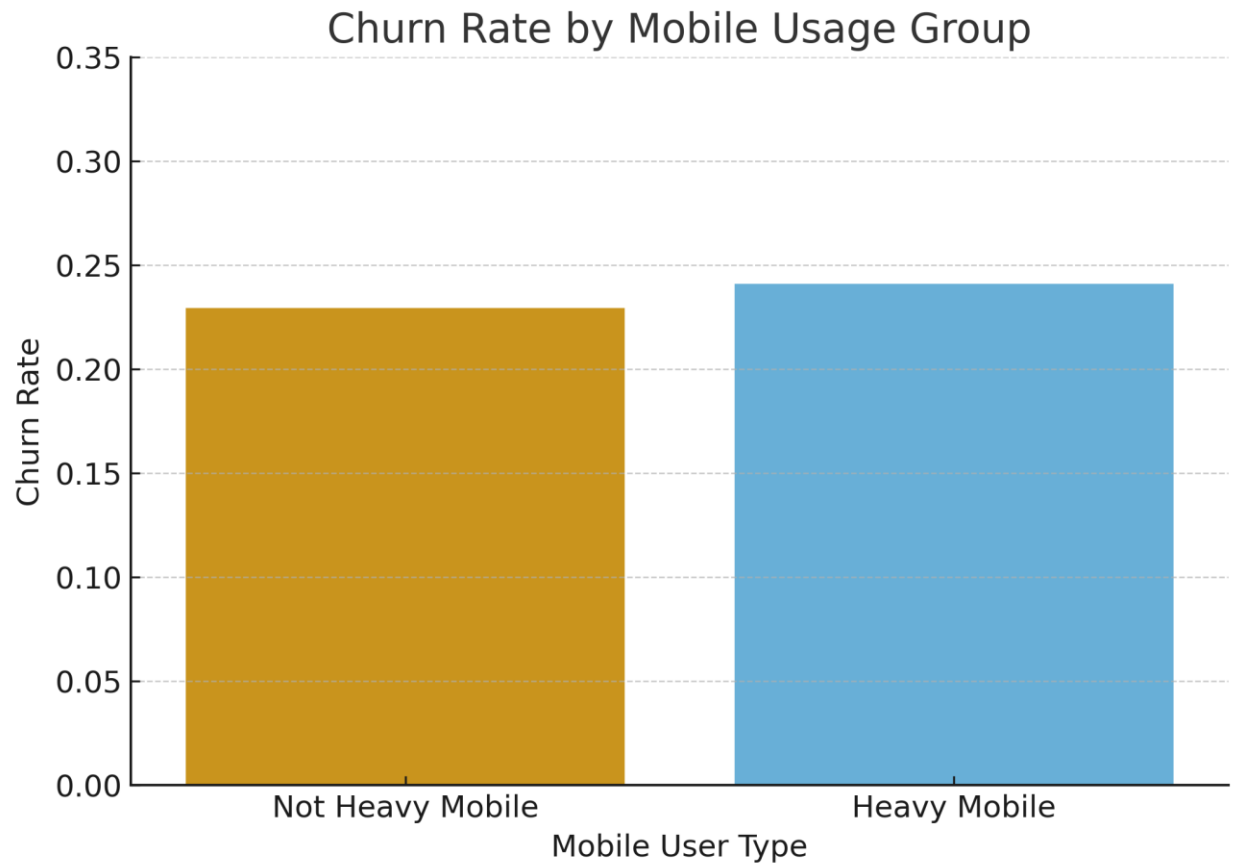
Business Questions



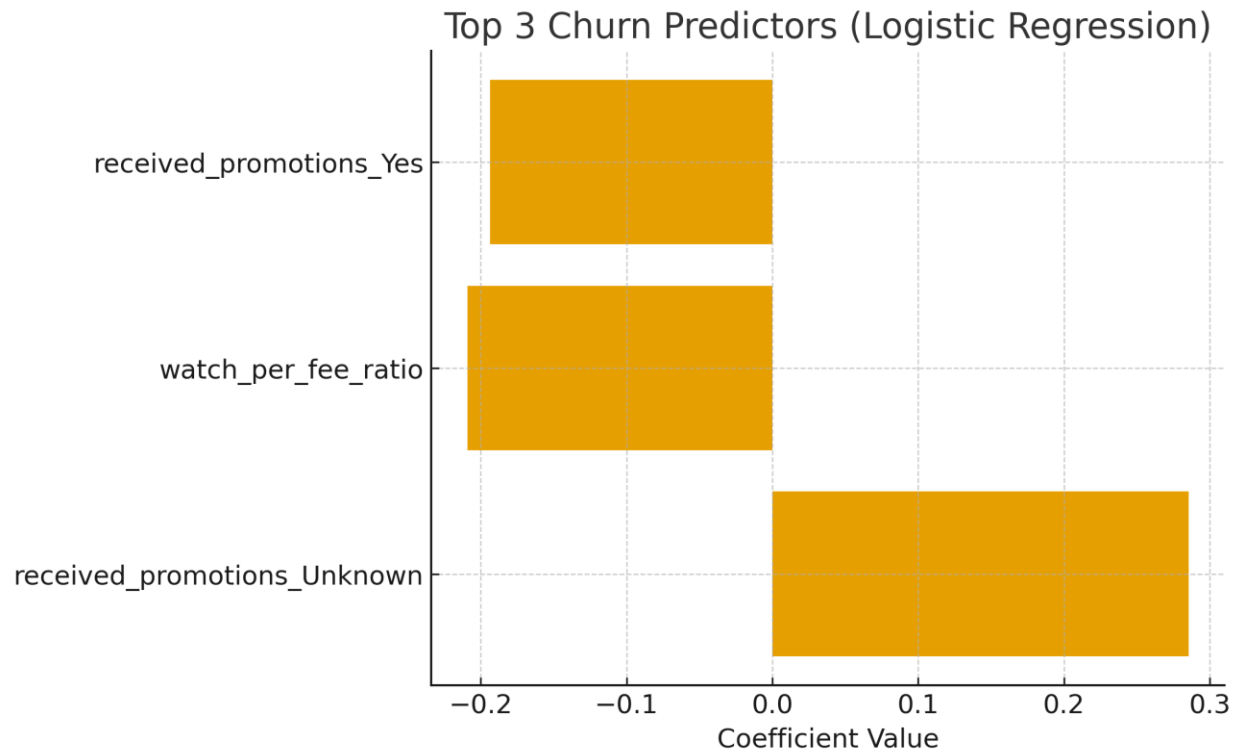
1. Users who receive promotions churn slightly less, but difference is not significant. This suggests promotions may help retention



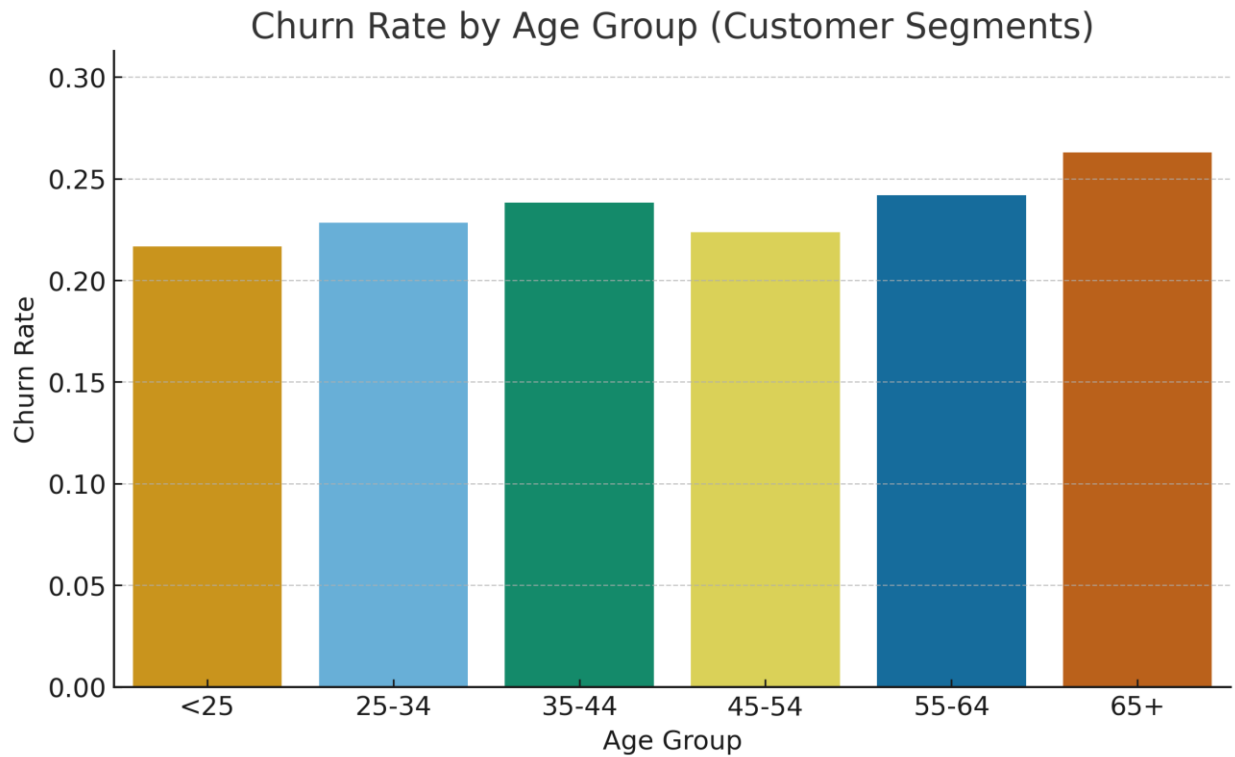
2. No. Churned users and active users have similar watch times, watch times does not significantly influence churn likelihood



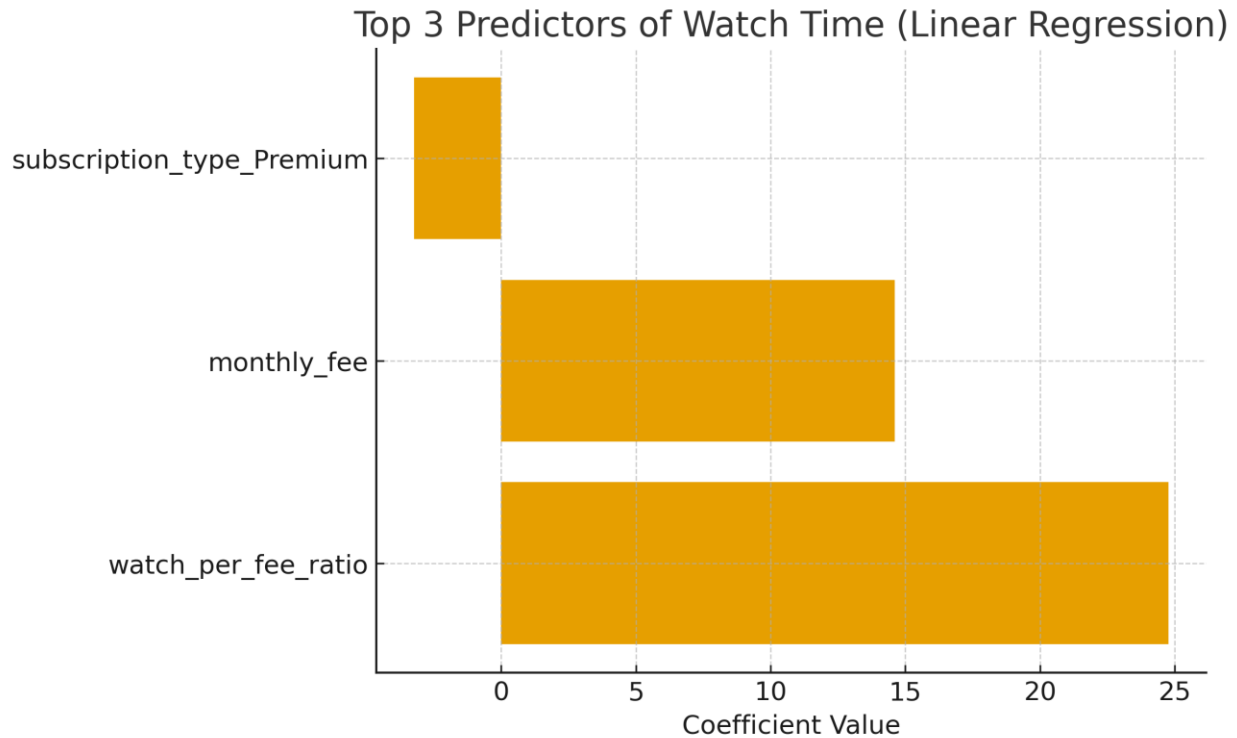
3. Mobile dominant users are not more likely to churn, being mainly a mobile user doesn't increase cancellation risk



4. Age 55+ show higher churn and are more likely to leave, watch_per_fee_ratio is lower churn. Users who get more viewing hours relative to cost churn less. Received promotions is lower churn.



5. Older users who churn more may need simple interface guidance and tailored content bundles. Low value users would need target content recommendations. Users not receiving promotions will need incentives to at-risk users who haven't received any offers



6. Heavy_mobile_user, tend to watch slightly more content. Some countries consistently watch more content. Watch time and tenure are not strongly predictable. More detailed behavior-based features would be needed for accurate modelling.

Recommendations

1. Improve value perception for at-risk users
 - Personalise content recommendations to increase watch time
 - Promote high-value or trending content to low-engagement users
2. Target older users with retention campaigns
 - Simplify onboarding flows
 - Promote content genres preferred by older demographics
 - Provide personalized help emails or tutorials

Data Issues or Risks

1. Imbalance (churn vs active users) the dataset contains more active users than churned users leading to imbalances. Insights about churn drivers may be weakened due to insufficient churn examples.
2. Limited behaviour data. Without signals, models performed poorly which limits the ability to identify true churn drivers
3. Weak predictive power, the dataset may not contain the variables that actually influence churn.